PROJECT PROPOSAL

Software Engineering Dissertation Project

-S.P. Navindu Prabhashana-

-CL/BSCSD/33/160-

Table of Contents

1.Title

BloxLogicAI: An AI and Blockchain-Enabled Supply Chain Forecasting Analysis System for Sri Lanka's Tea Industry.

2. Introduction

Sri Lanka's tea industry is a strategic foreign exchange earner and a key contributor to national GDP. Recent data show that tea export revenues exceeded USD 1.31 billion in 2023, reflecting the sector's enduring importance despite economic volatility (https://srilankateaboard.lk/wp-content/uploads/2025/04/English-AR-2023.pdf). However, stakeholders still face major challenges in predicting market demand, deciding monthly export volumes and managing disruptions such as fuel crises, geopolitical shocks and climate-related events.

Traditional decision-making in the tea export value chain often relies on descriptive statistics, expert judgement and spreadsheet-based trend analysis, which struggle to capture complex interactions among production, global demand, shipping constraints and macroeconomic conditions. At the same time, international buyers increasingly demand transparency and traceability regarding origin, sustainability, and ethical practices along the supply chain.

BloxLogicAI is proposed as an integrated AI and blockchain enabled supply chain forecasting and analysis system focused on Sri Lanka's tea sector. The system will:

- forecast market demand for key importing regions,
- predict how much tea should be exported next month and beyond,
- perform anomaly detection for shocks (e.g., war, fuel crisis, environmental disruptions) and
- provide blockchain-based traceability across the supply chain.

By combining predictive analytics with an immutable digital ledger, BloxLogicAI aims to strengthen resilience, transparency and decision quality for plantation companies, exporters, brokers, logistics providers and policymakers.

3. Current Situation and Problem Identification

Sri Lanka produced around 300 million kilograms of tea in 2022**,** with approximately 85% exported (Tea Exporters Association, 2024). Export volumes have fluctuated in recent years, with quantities falling from over 280–290 million kg per year in the late 2010s to around 239–247 thousand tonnes in 2022–2023 (https://www.cypherexim.com). Market reports from the Tea Exporters Association and Ceylon Tea Brokers show ongoing volatility in monthly export volumes and Free on Board (FOB) prices through 2023–2025 (Tea Exporters Association, 2024).

Key challenges in the current situation

1. Uncertain and fragmented demand forecasting

Exporters must decide shipment volumes to markets such as Russia, the Middle East, Europe and East Asia without robust statistical or AI-driven tools. Demand is affected by income levels, currency fluctuations, trade sanctions and consumer preferences that change over time.

2. Production variability due to environmental and input factors

Tea yields depend on rainfall patterns, temperature, incidence of pests and diseases and access to fertilizers. Climate variability and policy decisions on agricultural inputs can significantly alter production, as seen in other Sri Lankan crops during recent fertilizer policy shifts.

3. Exposure to external disruptions

The tea supply chain is vulnerable to disruptions such as global conflicts affecting major buyer countries, spikes in fuel prices influencing shipping costs and macroeconomic shocks linked to Sri Lanka's recent debt crisis. Current systems rarely provide early warnings for such anomalies.

4. Limited traceability and transparency

While tea is often branded as "Ceylon Tea", full traceability from estate to cup is not consistently visible to importers and consumers. Paper-based documentation and siloed databases make it difficult to verify origin and handling practices, reducing trust and limiting premium pricing opportunities.

5. Lack of integrated, real-time analytics

Existing reporting focuses on historical statistics rather than forward-looking insights. Stakeholders cannot easily simulate "what-if" scenarios such as a war affecting a major importer or a sudden fuel shortage.

Problem statement

Sri Lanka's tea industry lacks a unified, data-driven platform that combines AI-based demand and export forecasting, anomaly detection and blockchain traceability. This limits the ability of key stakeholders to plan monthly exports, anticipate disruptions and provide transparent supply chain information to global buyers.

4. Literature Review

AI and forecasting in agriculture and commodities

Forecasting agricultural production and commodity prices has been widely studied using a range of predictive models, from traditional statistical approaches to advanced machine learning techniques. While many studies explore deep learning models such as LSTM**,** ARIMA–LSTM hybrids and XGBoost**,** research also highlights the practicality and accuracy of simpler time-series forecasting tools such as Prophet**,** especially when data availability is limited.

Recent studies comparing machine learning and deep learning models reveal that advanced models like LSTM and XGBoost often outperform traditional statistical methods for highly nonlinear agricultural series (Ragunath & Rathipriya, 2025). However, Prophet has been shown to perform competitively and is valued for its simplicity, interpretability and strong ability to model seasonality, making it suitable for smaller datasets and operational environments typical in agricultural forecasting.

- Ragunath and colleagues compared ARIMA, SVR, Prophet, XGBoost, and LSTM for commodity price prediction and found LSTM models achieved lower error for highly nonlinear series (Ragunath and Rathipriya, 2025).

- Yuan et al. proposed an LSTM-based framework for crop price prediction and demonstrated its suitability for capturing seasonal and trend components in agricultural markets (Chen and Sin Kai Ling, 2020).

- Ray et al. introduced an ARIMA–LSTM hybrid model for volatile agricultural series, highlighting the benefit of combining statistical and deep learning techniques (Ray et al., 2023).

- Dionissopoulos et al. used AI models to forecast agricultural product and supply prices and emphasized how such forecasts support policy and farm-level planning (Dionissopoulos et al., 2024).

These studies collectively justify the use of Prophet in BloxLogicAI, as it provides a practical, low-complexity, yet accurate forecasting method appropriate for monthly export volume prediction and market demand analysis. While deep learning models appear frequently in literature, the chosen toolset prioritizes Prophet for its simplicity, efficiency, and suitability for undergraduate-level deployment.

Blockchain and agricultural supply chain traceability

Blockchain technology is increasingly used in food and agricultural supply chains to ensure product authenticity, traceability and transparency. Literature consistently identifies blockchain as a solution for addressing issues such as data tampering, fragmented record-keeping, counterfeit products and lack of trust between trading partners.

- Mirabelli & Solina (2020) highlight blockchain's value in improving transparency and traceability in agro-food chains.
- MS, R & M (2021) discuss benefits such as decentralization, smart contracts, and elimination of intermediaries in agricultural blockchain applications.
- Ellahi et al. (2023) provide systematic evidence that blockchain-based traceability improves trust, safety, and visibility from farm to consumer.
- Babu & Devarajan (2023) demonstrate blockchain + off-chain storage solutions (e.g., IPFS) to handle scalability while ensuring integrity.
- Chen et al. (2023) present a Hyperledger Fabric-based tea supply chain model capable of capturing detailed production records and preventing counterfeiting.

Although many studies use advanced permissioned networks such as Hyperledger Fabric, BloxLogicAI adopts a simplified blockchain simulation built in Python, using SHA-256 hashing to demonstrate immutability and traceability. This allows the system to align with academic literature while remaining achievable within the project's chosen toolset.

Anomaly detection in supply chains

Anomaly detection methods such as Isolation Forest, Autoencoders, and hybrid approaches are widely used to detect unusual patterns in high-dimensional data (Glaser, Harrison and Josephs, 2022).

- Glaser discusses applying various outlier detection algorithms, including Isolation Forest and Local Outlier Factor, to improve supply chain data quality.

- Almansoori et al. propose a hybrid Autoencoder–Isolation Forest framework to improve anomaly detection performance, especially where single models struggle (Mahmood Almansoori and Miklós Telek, 2023).

- Tutorials and practical implementations further illustrate Isolation Forest's strengths in fast, unsupervised anomaly detection (Digitalocean.com, 2024).

These techniques are well-suited to detecting abnormal shifts in export volumes, shipping times, production yields, or price indicators caused by shocks like war, fuel crises, or extreme weather.

Gap in existing solutions

Although significant research exists on:

- AI-based forecasting of agricultural commodities,
- blockchain-enabled supply chain traceability and
- anomaly detection frameworks,

there is no integrated system tailored specifically to Sri Lanka's tea export industry that simultaneously:

- predicts monthly market demand and export quantities using a tool like Prophet,
- detects anomalies using Isolation Forest and
- provides blockchain-backed traceability using an accessible Python-based ledger.

BloxLogicAI addresses this gap by combining these techniques into a unified, lightweight, and industry-focused solution.

BloxLogicAI is different and better, because it delivers a unique, integrated solution that the Sri Lankan tea industry currently lacks. Existing tools are fragmented, relying on manual spreadsheets, basic statistics or separate systems for forecasting, anomaly detection and traceability. In contrast, this project combines AI forecasting using Prophet, anomaly detection using Isolation Forest and blockchain-based traceability into one lightweight, affordable and industry-focused platform. It is specifically designed for Sri Lanka's tea export environment, addressing real issues such as export volatility, climate-driven production changes and limited transparency. By offering a practical, low-cost system that runs on simple hardware while improving decision-making, early-warning capability and supply chain trust, BloxLogicAI provides a clearly superior and much-needed solution.

5. Proposed Technique/Solution

BloxLogicAI is redesigned to run entirely on **Python**, using **Streamlit** for the UI, **Prophet** for forecasting, **Scikit-Learn**for anomaly detection and a **Python-based blockchain simulation** for traceability.

A. Functional Requirements – System Components

1. User Portal (Streamlit Web App)
Developed using Streamlit, with:

- Role-based login (basic Python session handling)
- Monthly demand forecast dashboards→ Implemented using Prophet
- Export quantity recommendation panel→ Simple rule-based + Prophet forecast
- Anomaly alerts→ Using Scikit-Learn Isolation Forest
- Blockchain traceability viewer→ Reading from locally stored blockchain ledger (Python hash chain)
- Data uploaded through CSV/Excel files→ Using Pandas

No complex backend frameworks: the entire app runs as a single Streamlit application.

2. Admin Portal (Streamlit Admin Dashboard)
Includes:

- User management (CSV-based storage or simple Python lists)
- Dataset upload and cleaning (Pandas)
- Model retraining triggers
- Blockchain ledger viewer (from Python hash chain)

No server clusters or container orchestration required.

3. AI Model (Python)

A. Demand Forecasting Module

- Implemented using Facebook Prophet
- Inputs: historical exports, country-wise demand datasets (CSV/Excel)
- Outputs: next-month and next-quarter predictions

B. Anomaly Detection

- Implemented using **Scikit-Learn Isolation Forest**
- Detects abnormal:
    - drops in export volumes
    - spikes in shipping durations
    - deviations in production vs climate

C. Export Quantity Recommendation

- Simple Python logic:
    - Forecasted Demand - Current Stock + Safety Margin

No deep learning models or GPU requirements.

4. Blockchain Module

A simple blockchain implemented entirely in Python:

- Each block hashed using **SHA-256**
- Stores tea batch events:
    - harvesting
    - processing
    - blending
    - packaging
    - export booking
- Ledger stored in JSON files or CSV
- QR codes generated using a Python library (optional)

This replaces Hyperledger Fabric with a simple, demonstrable prototype.

5.1 Future upgrades

Future upgrades include moving from CSV files to a real database, implementing a full Hyperledger Fabric blockchain network, developing a mobile application, integrating real-time data APIs, adding IoT sensor-based tracking, adopting deep learning forecasting models, providing automated export documentation, supporting multiple languages, enabling supply chain simulation and deploying the system on cloud infrastructure for scalability.

B. Non-Functional Requirements

Adjusted to simpler tech:

- Security: local encryption + hashed block records
- Performance: Prophet and Isolation Forest run quickly on CPU
- Usability: fully powered through Streamlit dashboards
- Reliability: ledger stored as append-only JSON file

6. Feasibility Analysis

Technical Feasibility

Tools used are lightweight and beginner-friendly:

- Python (core development)
- Prophet (forecasting)
- Scikit-Learn (anomaly detection)
- Pandas (data handling)
- Streamlit (UI)
- Local Python blockchain script (SHA-256 hashing)
- CSV/Excel files

Economic Feasibility

Costs remain near zero because:

- All tools are free and open-source
- No servers, nodes, containers, or cloud GPUs are required

Operational Feasibility

Stakeholders can easily use CSV upload and Streamlit dashboards.

7. Project Description

In summary, BloxLogicAI will:

- A Streamlit web application for users/admins
- Forecasting with Prophet
- Anomaly detection with Isolation Forest
- Lightweight Python blockchain for traceability
- Data stored and loaded using CSV/Excel files

8. Deliverables

1. Streamlit User Interface
2. Streamlit Admin Panel
3. Prophet-based forecasting model
4. Isolation Forest anomaly model
5. Python blockchain simulation
6. All datasets stored in CSV/Excel
7. GitHub repository
8. Technical documentation + diagrams
9. Final report & presentation

9. Resources Required

Software / Tools

- Python (core development language)
- Prophet (forecasting)
- Scikit-Learn (Isolation Forest)
- Streamlit (web UI)
- Pandas (data processing)
- CSV / Excel files (primary storage)
- Python Blockchain Simulation Script (SHA-256 ledger)
- Git / GitHub (version control)

Hardware

- Standard development laptop (no GPU required)

Data sources

- Sri Lanka Tea Board annual reports and statistics (https://srilankateaboard.lk/wp-content/uploads/2025/04/English-AR-2023.pdf).
- Government/industry data books for tea production and export volumes (https://www.industry.gov.lk/web/wp-content/uploads/2025/04/Tea-2023.pdf).
- Market reports from industry associations and brokers (https://teasrilanka.org/market-reports).
- Weather and climate data from meteorological sources.
- Macro-economic, fuel price and geopolitical indicators (e.g., news feeds, economic data services).

10. Expected Output and Outcome

Expected Output (Tangible)

- A working web-based system (BloxLogicAI) accessible to users.
- Implemented AI models with documented accuracy metrics.
- A running Hyperledger network with sample nodes and traceable tea batches.
- Visual dashboards for forecasts, anomalies, and supply chain traces.

Expected Outcome (Impact)

- Improved forecasting: more accurate estimation of monthly export volumes and market demand, leading to better resource planning.
- Enhanced resilience: earlier detection of crises (e.g., war, fuel issues, climate anomalies) affecting the tea supply chain.
- Higher transparency and trust: blockchain-backed traceability improves confidence among overseas buyers, supporting premium pricing and brand value.
- Better policy support: aggregate analytics can support national-level planning and export strategies.

11. Time Plan for Implementation

Week	Tasks / Activities
Week 1 – Week 2	Requirement analysis, finalize system scope, gather datasets, define architecture, prepare diagrams
Week 3 – Week 4	Dataset collection, cleaning & preprocessing using Pandas, handle missing data, structure data for AI & blockchain modules
Week 5 – Week 6	Develop Prophet forecasting model, train & evaluate predictions, build Isolation Forest anomaly detection model
Week 7 – Week 8	Develop Python-based blockchain module, implement SHA-256 hashing, create ledger structure, test batch traceability
Week 9 – Week 10	Build Streamlit User & Admin Portals, integrate dashboards, implement dataset upload, model retraining, traceability viewer
Week 11	System integration, connect models with UI, connect blockchain with UI, complete workflow debugging
Week 12	Functional testing, performance testing, accuracy evaluation, user experience improvement
Week 13	Write documentation, user manual, technical documentation, diagrams, GitHub documentation
Week 14	Final report preparation, presentation slides, rehearsal, final system demo, submit project

12. Limitations, Risks, and Critical Evaluation

Limitations

- Data dependency: Model performance heavily depends on the quality and granularity of historical tea production, export, and climate data. Missing or inconsistent records can reduce accuracy.
- Model generalisation: Forecasts may not fully capture unprecedented events (e.g., sudden geopolitical conflicts) despite anomaly detection (Glaser, Harrison and Josephs, 2022).
- Blockchain overhead: Permissioned blockchain networks may have scalability and latency constraints; careful design is required to ensure acceptable performance (MS, R and M, 2021).

Risks

- Cybersecurity risk: As a web-based system, BloxLogicAI is exposed to cyber threats; robust authentication, authorization, and monitoring are required.
- Adoption risk: Smaller stakeholders may initially resist new digital workflows or perceive blockchain as complex.
- Model drift: Over time, structural changes in the market or policy environment may require retraining and revalidation of models.

Critical Evaluation Approach

- Regularly evaluate model performance using rolling forecasts and back-testing (Dionissopoulos et al., 2024).
- Perform usability testing with representative users (exporters, plantation managers).
- Conduct post-implementation reviews focusing on whether forecasts and anomaly alerts led to better decisions.
- Compare blockchain-based traceability against existing documentation flows to quantify improvements in transparency.

13. References

1. Babu, S. and Devarajan, H. (2023). Agro-Food Supply Chain Traceability using Blockchain and IPFS. *International Journal of Advanced Computer Science and Applications*, 14(1). doi:https://doi.org/10.14569/ijacsa.2023.0140142.

2. Chen, C.-L., Zhan, W.-B., Huang, D.-C., Liu, L.-C., Deng, Y.-Y. and Kuo, C.-G. (2023). Hyperledger Fabric-Based Tea Supply Chain Production Data Traceable Scheme. *Sustainability*, [online] 15(18), p.13738. doi:https://doi.org/10.3390/su151813738.

3. Chen, Z. and Sin Kai Ling (2020). Long Short-Term Memory Model Based Agriculture Commodity Price Prediction Application. doi:https://doi.org/10.1145/3417473.3417481.

4. Digitalocean.com. (2024). *Anomaly Detection Using Isolation Forest in Python | DigitalOcean*. [online] Available at: https://www.digitalocean.com/community/tutorials/anomaly-detection-isolation-forest.

5. Dionissopoulos, I., Assimakopoulos, F., Spiliotopoulos, D., Margaris, D. and Vassilakis, C. (2024). Predicting Agricultural Product and Supplies Prices Using Artificial Intelligence. *Proceedings of the 16th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management*, [online] pp.371–379. doi:https://doi.org/10.5220/0013071600003838.

6. Ellahi, R.M., Wood, L.C. and Bekhit, A.E.-D.A. (2023). Blockchain-Based Frameworks for Food Traceability: A Systematic Review. *Foods*, [online] 12(16), p.3026. doi:https://doi.org/10.3390/foods12163026.

7. Glaser, A., Harrison, J. and Josephs, D. (2022). Anomaly Detection Methods to Improve Supply Chain Data Quality and Operations. *SMU Data Science Review*, [online] 6(1), p.3. Available at: https://scholar.smu.edu/cgi/viewcontent.cgi?article=1211&context=datasciencereview.

8. Mahmood Almansoori and Miklós Telek (2023). Anomaly Detection using combination of Autoencoder and Isolation Forest. pp.25–30. doi:https://doi.org/10.3311/wins2023-005.

9. Mirabelli, G. and Solina, V. (2020). Blockchain and agricultural supply chains traceability: research trends and future challenges. *Procedia Manufacturing*, 42, pp.414–421. doi:https://doi.org/10.1016/j.promfg.2020.02.054.

10. MS, H., R, S. and M, R. (2021). Block Chain Based Agricultural Supply Chain-A Review. *Global Transitions Proceedings*. doi:https://doi.org/10.1016/j.gltp.2021.08.041.

11. Ragunath, R. and Rathipriya, R. (2025). Predicting Agriculture Commodity Price Trends Using Ensemble Regression Approach. *Journal of Agricultural Sciences – Sri Lanka*, 20(3). doi:https://doi.org/10.4038/jas.v20i3.10534.

12. Ray, S., Lama, A., Mishra, P., Biswas, T., Sankar Das, S. and Gurung, B. (2023). An ARIMA-LSTM model for predicting volatile agricultural price series with random forest technique Image 1. *Applied Soft Computing*, 149, p.110939. doi:https://doi.org/10.1016/j.asoc.2023.110939.

13. Tea Exporters Association. (2024). *Tea Exporters Association Sri Lanka*. Available at: https://teasrilanka.org/market-reports.